

Polydimethyl siloxane coatings with superior antibiofouling efficiency in laboratory and marine conditions



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ABSTRACT

Aim of this investigation was to demonstrate in laboratory and field experiments, the ability of relevantly selected non-ionic surfactant, non-toxic antioxidant and their combinations to reduce sharply marine biofilm development on polysiloxane coatings totally preventing marine macro-fouling on slowly moving and static immersed surfaces.

Their superior antifouling properties: reduced to zero mussel and algae adhesion, as well as multi-species macro-fouling, and 6–10 folds reduced marine biofilm, were attributed to a formation of a thin, plane, "bioactive" surface layer.

Its excellent anti-biofouling action was supposed to be due to the presence of impeding the initial microbial attachment non-ionic surfactant that together with the non-toxic antioxidant inhibits a supposed oxidative cross-linking of secreted exopolymeric substances and later assists the biofilm dispersal.

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1. Introduction

The marine biofouling is a natural phenomenon affecting any submerged in water material surface with many negative issues, such as extra energy consumption at vessels, increased corrosion, negative ecological issues and other with very high overall economic impact [1]. Since 2008, when the effective but environments damaging tributyltin paints were banned, siloxane fouling-release coatings became a viable commercial alternative to the toxic biocide antifouling paints.

Variety of functional polymer coatings and brushes were developed so far with less environmental impact than traditional biocides [2]. Hydrophilic polymers, such as polyethylene glycol, zwitterionic polymers and polysaccharides, were found to resist effectively the attachment of marine organisms due to their extensive hydration, but they appeared to be not enough stable in water environment. Polycationic coatings were effective in reducing of

marine biofouling partly because of their good bactericidal properties [2].

New strategies for a controlled functionalization of surfaces by environmentally benign non-biocidal sol-gel antifouling and foul-releasing coatings were investigated [3]. Variety of amphiphilic triblock copolymers with PEGylated hydrocarbon structures were proposed for this aim [4] as well as a rational strategy to control the surface segregation in dip-coating by the employment of a low-molecular-weight surfactant [5].

Fouling-release coatings containing surface-active polymers derived from styrene monomers showed that the surface-active polymer preferentially populates the outermost layer of the coating, despite its low content in the blend. The self-segregation process and the response to the external environment were strongly dependent on both, the chemistry of the surface active polymer and the type of matrix elastomer [6]. Developed by Sun et al. [7], perfluoropolyether/polydimethyl siloxane/acrylic polyols based coatings demonstrated an adhesion and antifouling properties that were comparable to a silicone standard.

The attempts to decrease marine biofouling by using nanoparticles [8], natural products, their modified synthetic analogues, some mixtures [9–12] and quorum sensing inhibitors [13] continue. A

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potential cytotoxicity, aggregation of the nanoparticles, too low ratio extracted active substance/natural raw material and other limits their practical application.

Elastomer coatings prepared by improving the original silicone formulations of Milne [14,15] were already accepted as a viable commercial non-toxic alternative of the toxic biocide paints. Special textures of the polydimethylsiloxane (PDMS) coatings were proved to enhance their release ability but insufficient ability to resist the exploitation conditions also limited their practical application [16]. Fouling-release coatings, based on poly(dimethylsiloxane) elastomers and fluoro-polymers minimized the adhesion between marine organisms and material surfaces leading to easy removal of biofoulers but they worked effectively only at high speed moving vessels [17,18].

Technologies, effective against hard shell and weed growth are already known but they continue to struggle against microorganisms slime build up [19,20]. Current study of a long-term micro-fouling on commercial biocide coatings, demonstrated once again that marine biofilm develops even on them [21].

Antifouling coatings were found to influence the structure of the community and abundance of colonizing biofilms, depending on both the coating type and the biofouling species [22]. The hydrodynamic stress was found to influence significantly the micro-fouling community, and irrespectively differences were observed in the fouling community developed under static and dynamic conditions [23]. But no one coating is known so far that is able to stop the development neither of biofilm under dynamic either less under static conditions.

The known approaches to reduction the biofilm development are targeted at: (i) creation of anti-adhesive material surfaces don't allowing microbial attachment and (ii) disturbance of any stage in the biofilm development by employment of anti-biofilm agents, like surfactants/bisurfactants, quorum sensing (QS) inhibitors and/or inhibitors of the cross-linking of exopolymeric substances (EPS) [24].

The versatile nature of the secreted by the biofoulers EPS and their ability to adsorb by different mechanisms in front of complementary in nature surfaces make exclusively difficult the complete prevention of their reversible adhesion. Therefore further release of anti-biofilm agent is necessary to restore the biofilm formation, i.e. the more effective non-toxic biofouling control requires creation of low adhesive surfaces that in addition possess delivery of non-toxic antibiofilm agents. The low adhesive surface was expected to impede the initial attachment of the planktonic cells and the anti-biofilm agents to restore some stage(s) in the development of biofilm [24].

This motivated us to try to improve the anti-biofouling ability of earlier developed low adhesive siloxane antifouling coatings totally preventing macro-fouling of high and low speed moving vessels and even of static immersed underwater constructions [20] by including of some low/nontoxic anti-biofouling agents in their composition. Our idea was to employ (i) low toxic surfactant contributing to the impeding of the initial adhering of approaching planktonic cells and/or inhibitor of the crosslinking of secreted EPS.

More effective inhibition of microbial adhesion by some non-ionic surfactants as compared to ionic ones was found by Whitekettle [25] and Cloete et al. [26]. Therefore, a non-ionic surfactant was preferred for this investigation.

According to the hypothesis of Mayer' et al. [27] surfactants' elutes, presenting on/or at near the interface are able selectively to inhibit the sulphhydryl cross-linking of bioadhesives. In addition, supposing possible oxidative cross-linking of microbial EPSs [28–30], a non-toxic antioxidant was employed for this investigation as a second anti-biofilm agent with the expectation to inhibit the crosslinking of secreted EPS.

The aim of this study was to demonstrate experimentally, under laboratory and field conditions, the anti-biofouling efficiency of relevantly selected non-ionic surfactant, non-toxic antioxidant and their combinations in biocide-free low adhesive siloxane composition coatings.

2. Materials and methods

2.1. Composition coatings

Composition coatings, based on combination of room temperature vulcanizing (RTV) siloxane elastomers (Gelest, USA), siloxane oil (DMS-T31, Gelest, USA), crosslinking agent (PSI-021, Gelest, USA), catalyst (SND 3260, Gelest, USA), and modifying agent (non-ionic surfactant, or/and non-toxic antioxidant), prepared as described in [20] were used for this investigation.

2.2. Modifying agents

Two modifying agents: non-ionic surfactant (NIS), alkoxyated trisiloxane (Super spreader Y17112, Momentive Comp, USA) and non-toxic antioxidant (NAO), α -tocopherol (E307, food antioxidant, Panteley Toshev Ltd., Sofia, Bulgaria) were employed for this study. They were added to the coating composition separately or in double combinations at weight ratio of 1:3 and 1:1. If not otherwise stated all chemicals were from Alfa Aesar Comp., Karlsruhe, Germany, p.a. quality.

2.3. Coated test-samples

Glass plates ($100 \times 100 \times 2$ mm and $10 \times 10 \times 2$ mm) were spin coated (at 400 min^{-1}) with primer, consisting of ethyltriacetoxysilane (50 wt.% toluene solution) and catalyst (3 wt.% dibutyltin dilaurate) to provide good adhesion of the coating to the glass surface. The primed dry glass plate was then covered with a corresponding composition by spin coating under the same conditions. In this way prepared test-samples were kept at room conditions for 30 days before the immersion in artificial seawater aquarium (ASW) or in Black sea, Yacht harbor, Nessebur, Bulgaria. The dry coating thickness was of 220–240 μm , measured by Stereomicroscope Leica MZ16 FA (Leica, Wetzlar, Germany).

2.4. Elastic modulus (Et)

The elastic modulus (Et) of the cross-linked polysiloxane compositions was estimated according to ISO R37 using samples with thickness of 2 mm.

2.5. Depth sensing indentation (DSI)

Dynamic Ultramicro-Hardness Meter DUH-211 S (Shimatzu, Japan) was employed for evaluation of the indentation hardness (HIT) and indentation elastic modulus (EIT) under the following conditions: test force 0.30 mN; loading speed 6.0 (0.0250) mN/sek; depth of penetration 20 nm.

2.6. Atomic force microscopy (AFM)

Easyscan 2 ("Nanosurf", Swiss) apparatus equipped with Pointprobe Contr-10 silicone SPM sensor (dimensions of $2 \times 450 \times 50 \mu\text{m}$) was employed to obtain plane and 3D images of the studied dry surfaces, operating in contact mode. Diamond Viker's pyramid with pike angle of 136° was used for all measurements at room temperature with loading speed of 0,0250 mN/s.

2.7. Contact angle (θ) and surface energy (E)

Contact angle measuring instrument Easy Drop (Kruss, Germany) was employed for contact angle measurement (angle resolution $\pm 0.1^\circ$) using three liquids with known surface tension: water, ethylene glycol and *n*-hexadecane. Surface energy (E) was calculated according to Fowkes' method [31].

2.8. X-Ray photoelectron spectroscopy (XPS)

XPS analysis was carried out with an ESCALAB II MK VG spectrometer (Thermo Scientific, UK) with AlK_{α} excitation X-ray source (excitation energy 1486.6 eV). Spectral scans and detailed recordings of the main peaks were made at 10^{-8} Pa. The binding energy scale was fixed by assigning $E_b = 285$ eV to the $(-CH_2-)$ C1s peak [32].

2.9. Mussel adhesion

All samples with dimensions of $100 \times 100 \times 2$ mm were put into a plastic bucket with every day changed demineralized water over a period of 7 days. For a test run, two samples were placed in an aquarium and 4 mussels of the approximately same size were placed posterior and anterior sides switched and with the dorsal side upside. The mussels were not fixed, so that during the time they moved around on the surface or left the surface.

2.10. Algal biofilm grown

Glass plates, treated like above described, have been half-sided coated. The plates were stored for seven days in artificial seawater (ASW) before placing in the aquarium. After 12 weeks in the aquarium the plates were washed with ASW and distilled water. After drying the plates were analyzed with fluorescent microscope Axiomager M.1 (Zeiss, Jena, Germany) using filter set 9 (488 nm). All samples were stained with SYTO 9 0.6 μ L/mL NaCl. No fluorescence dye was used because the auto fluorescence of the slime algae biofilm showed strong intensities. Pictures were taken with a digital camera AxioCam MR3 (Zeiss, Jena, Germany).

2.11. Bacterial adhesion

Bacterial strain used was an overnight culture of *Marinobacter hydrocarbonoclasticus* (*M. hydrocarbonoclasticus*, DSM 50418). The optical density was adjusted to 0.1 at 600 nm, which corresponds to 4×10^7 cfu/mL using a UV/Vis Spectroscopy device (Specord 200, Analytik Jena, Germany). The test samples ($10 \times 10 \times 2$ mm) were exposed to the bacterial solution for 1 h and 4 h. The bacterial adhesion was analyzed with a scanning device, Mithras LB 940 (Berthold Technologies Ltd., Germany) according to the protocol described by Pranzetti et al.. The samples were first exposed to ASW for one hour on a plate shaker at 50 rpm. Afterwards they were put in contact with a minimal medium solution of Marine Broth (Carl Roth GmbH, Karlsruhe, Germany) and ASW 1:100 (containing 4×10^7 cfu/mL of *M. hydrocarbonoclasticus*) at room temperature, on a plate shaker (50 rpm) for the respective time of analysis. The samples were then rinsed with ASW, fixated with glutaraldehyd 2.5% in ASW, rinsed once again with ASW, followed by a staining with SYTO 9 (green staining) from the LIVE/DEAD BacLight Bacterial Viability Kit (Molecular Probes, Eugene, Oregon, USA) and observed with a fluorescence microscope (Zeiss Axio Imager M1, Carl Zeiss, Jena, Germany). The pictures of the samples followed by this step are listed according to the incubation time.

2.12. Long-term marine biofouling and biofilm characterization

Long-term (one year) sea exposure was carried out in Black sea, Yacht harbor, Nessebur, Bulgaria for the period April 2014–April 2015. The samples were fixed on a common plane plate and vertically immersed at a depth of 3 m for 1 year. 5 samples of each coating were used for long-term micro-fouling evaluation. The biofilm of all 5 samples was carefully removed by means of a plastic spatula and placed in 15 ml Falcon type test-tube. 5 ml distilled water was added to each test-tube. Homogenization of the biofilm mass was performed for 5 min, directly in the test-tube by dipping of an ultrasound rood (Bandelin electronic UW 3100) and cooling in ice bath.

2.13. Total chlorophyll (A and B) and carotenoids (carotenes and xanthophylls)

The total chlorophyll and carotenoids were spectrometrically estimated from ethanol extract, employing Pharmacia Biotech Ultraspec 3000, Germany apparatus. The pigments extraction from the mass of the biofilm was performed as follows: 1 ml from the homogenized biofilm suspension was carefully, drop-way placed onto glass filter (pore size of 1.6 μ m). The reach of pigments cell elements were held on the filter and each filter was placed in 2 ml Eppendorf type centrifuge test-tube. 1.5 ml 95% ethanol was added and the containers were dark-kept for 4 h under mixing on KIKA KS250 (Labortechnik shaker, Germany). The complete extraction was followed by 5 min centrifuging at $28,000 \times g$ for clearing of the extract. 1 ml from each clear extract was used to calculate the content of: carotenoids – from spectrometry at 470 nm; chlorophyll B – at 649 nm; chlorophyll A – at 664 nm; for turbidity estimation and the other absorbance's correction – at 730 nm with an UV-vis spectrophotometer (Specord 200, Analytik Jena, Germany).

2.14. Biofilm dry mass

Biofilm dry mass was estimated gravimetrically as follows: 3 ml from each suspension were placed in preliminary weight (± 0.0001 g) Falcon test-tube (dried for 24 h at 105°C and conditioned for 4 h at normal conditions) and centrifuged at $5200 \times g$ for 5 min. The liquid phase was separated and 5 ml isopropanol (99.7%) was added to remove the residual siloxane oil from the sediment. After centrifuging and separation of the isopropanol, the test-tube with the sediment was dried for 5 h at 90°C , conditioned for 4 h at normal conditions and weight (± 0.0001 g). The biofilm dry mass was estimated as difference between this weight and initial weight of the empty test-tube.

3. Results

3.1. Surface characteristics and elastic modulus (E_t)

The main surface characteristics influencing bioadhesion and biofouling of the studied biocids-free composition coatings, estimated before immersion in seawater are presented in Table 1. It is evident that the non-ionic surfactant (NIS) and non-toxic antioxidant (NAO) slightly influence the water contact angle (θ_{H_2O}) and surface energy (E). All coatings are strong hydrophobic demonstrating water contact angle (θ_{H_2O}) higher than or around 90° , slightly varying on the nature and amount of the additive. Having surface energy (E) of 25.82–26.71 mNm, all studied coatings fall within the so called “Baier's window” accepted as the best for good biofouling release. In addition, all coatings were plane, except the one containing both, NIS and NAO (sample 4, Table 1). It demonstrated some small waylays creating insignificant nanoscale surface roughness (R_a of 15.914 nm and R_q of 20.211 nm). The last one is far below the size of the test bacterium, *Marinobacter hydrocarbonoclasticus* (gram negative, rood shaped; on average between 0.3–0.6 μ m

Table 1

Surface characteristics, water contact angle (WCA), surface energy (E), indentation elastic modulus (EIT), indentation hardness (HIT), dynamic hardness (HMT), surface roughness (R_a and R_q) and elastic modulus (Et) of the studied coatings: 1 – control (without additives); 2 – containing 0.5 wt.% non-ionic surfactant (NIS); 3 – 1.5 wt.% antioxidant (AO); 4–0.5 wt.% NIS + 1.5 wt.% AO.

Sample Parameter	1 Control	2 NIS (0.5 wt.%)	3 AO (1.5 wt.%)	4 NIS + AO (0.5 wt.% + 1.5 wt.%)
WCA [°]	102.2	89.9	99.2	98.6
E [mNm]	25.82	25.87	26.43	26.71
R_a [nm]	Plane surface	Plane surface	Plane surface	15.914
R_q [nm]				20.211
EIT [N/mm ²]	1.252e	9.582e	1.123e	9.607e
HIT [N/mm ²]	0.247	0.156	0.214	0.182
HMT [N/mm ²]	0.095	0.068	0.084	0.072
Et [Pa]	318	316	298	280

in diameter and 2–3 μm long, Wikipedia). The very low indentation elastic modulus, EIT (of 1.125–9.607 N/mm²) that is of several orders (about 10^5) below the elastic modulus, Et (Table 1) indicated the presence of a thin, very weak surface layer, with slight differences in the surface resistance to plastic deformation (as indicated by HIT, Table 1) and total deformation (as indicated by HMT, Table 1). This layer could be formed by leached from the coating liquid components, namely oil, oil/NIS, oil/NAO or oil/NIS and NAO. Outside of the rubber matrix, the NIS and NAO were unlimited miscible, forming a true solution whereas the NIS and oil were immiscible. Thus, one could expect a thin oil layer presence on the surface of the control (Table 1, sample 1); oil/NIS layer on the sample containing NIS, since the oil and NIS are immiscible (Table 1, sample 2); oil/NAO if NAO is the modifying agent (Table 1, sample 3); or oil/NIS-NAO solution on the coating containing both, NIS and NAO since NIS and NAO were unlimited miscible (Table 1, sample 4). The slight differences among the surface characteristics of the studied coatings (Table 1, samples 1–4) could be ascribed to different composition of this surface layer.

The high surface hydrophobicity of all plane or almost plane surfaces combined with a low elastic modulus can be a reason for the very good antifouling properties of the studied biocides-free coatings. The elastic modulus (Et) of the coatings containing NAO of about 298–280 Pa (Table 1, samples 3 and 4) was slightly decreased as compared to that of the control coating of 318 Pa (Table 1, sample 1), that indicated some differences in the vulcanization network and in this way possible participation of the NAO in its formation.

3.2. Mussel adhesion

Due to a known efficacy of some non-ionic surfactants as inhibitors of biological adhesion and an assumed oxidative crosslinking of secreted EPS [28–30,33] from marine biofoulers, a nonionic surfactant, NIS and a non-toxic antioxidant, NAO were selected for this investigation. Several test surfaces were used to be evaluated mussel adhesion: bare glass as well as glass covered with: control coating without additives (Table 1, sample 1); coating containing NIS (Table 1, sample 2) or NAO (Table 1, sample 3) or both, NIS and NAO (Table 1, sample 4). It was observed that attachment happens on the bare glass samples as well as on those covered with (i) control coating without additives (Table 1, sample 1) or (ii) containing NIS (Table 1, sample 2). More than three byssus plaques were needed to hold the mussel constant on the surface. The secreted plaques on the bare glass were much stronger attached as compared to those on the covered surfaces. The mussels did not detach under their own weight from the bare glass samples when taken out of the aquarium, whereas the mussels attached to the surfaces covered with both: coating without additives or containing NIS, dropped freely under their own weight in about 5 s. It was observed that on an average, the attachment on the surfaces occurs similar often to those coatings: control without additives (Table 1, sample 1) and coating containing NIS (Table 1, sample 2).

If the mussel was fixed on the surface (Fig. 1), the plaque attachment count was similar for the samples covered with both: coating without additives (Fig. 1a; Table 1, sample 1) or containing NIS (Fig. 1b; Table 1, sample 3), but the produced plaques in total were much higher on the coating without additives (Fig. 1a). Decreasing the plaques production, NIS decreased adhesion strength on the interface mussel/coating and thus assisted additionally the releasing ability of the coatings. In addition, the mussels did not attach at all to the samples covered with coating containing NAO (Table 1, sample 3) or both, NIS and NAO (Table 1, sample 4).

A comparison of the mussel adhesion on bare glass to that of covered glass samples confirmed the well-known ability of the low surface energy siloxane coatings to decrease mussel adhesion. The experimental results demonstrated also the ability of the selected NAO to inhibit the crosslinking of mussel EPSs, the effect being an indirect conformation for oxidative mechanism of this process.

3.3. Algae biofilm grown

To look whether an algae biofilm develops on the studied surfaces, some samples were left in the aquarium for 12 months. Fluorescence images of the formed biofilms are presented in Fig. 2. Algal biofilm growth was significant on the bare glass slides (Fig. 2a), whereas no algal biofilm grown was observed on the glass slides covered by coating containing NIS (Fig. 2b). Evidently the selected NIS was able to stop totally the algae biofilm grown.

3.4. Bacterial adhesion

Marinobacter hydrocarbonclasticus adhesion was tested on all covered samples as well as on bare glass and the results are presented in Fig. 3. Generally, the *Marinobacter hydrocarbonclasticus* had no problems with the adhesion on all studied surfaces but some slight differences were observed among the bare glass and samples covered with the studied coatings. A comparison of the left column of pictures in Fig. 3(a, c, e, g, i) showed the differences in the bacterial films developed after 1 h exposure in ASW: confluent biofilm was observed on the bare glass (Fig. 3a) whereas that on the sample covered with coating containing both, NIS and NAO (Fig. 3i) was scarce. The right column of pictures in Fig. 3(b, d, f, h, j) demonstrate bacterial colonies formed on the same samples after 4 h exposure in ASW. It is evident, that the biofilm dispersal was most expressed on the coating containing NIS and AO (Fig. 4j).

The effects of NIS, NAO and their combinations were slightly pronounced at mono-specie bacterial film. On contrary, the mussel and algae adhesion were sharply decreased in the presence of the same surfactant, antioxidant and their combinations. This is not a surprise, taking into account the different composition of the bacterium, mussel and algae EPS as well as their different adsorption and crosslinking mechanisms.

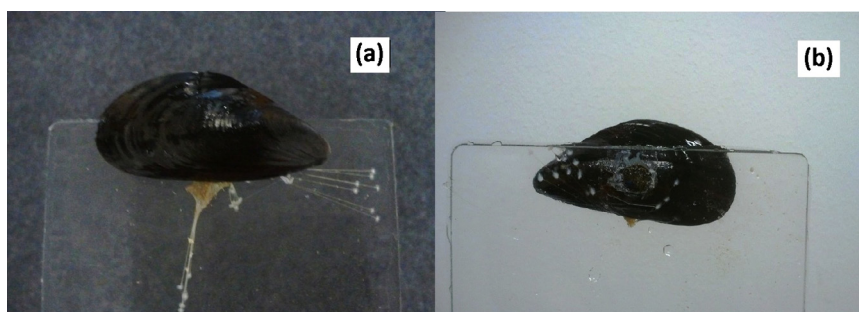


Fig. 1. Plaques produced on: (a) – control coating without modifying agents, sample 1; (b) – coating containing 0.5 wt.% non-ionic surfactant, sample 2.

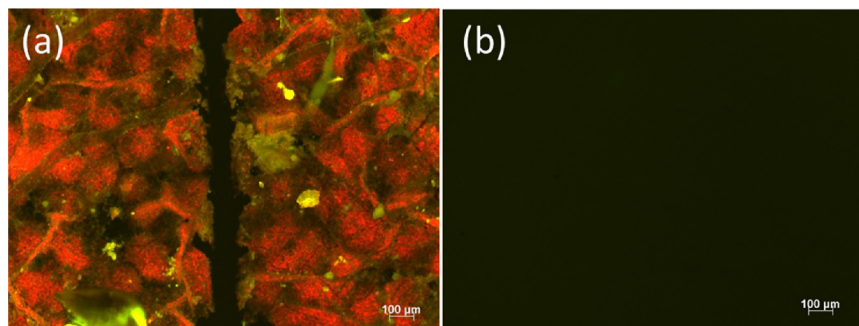


Fig. 2. Fluorescent images (magnification 50 $\times$) of algae biofilm grown for 12 weeks on: (a) – bare glass scratch in the middle; (b) – coating containing 1.0 wt.% NIS.

3.5. Sea biofouling and multi-species biofilms formation

No macrofoulers, but scarce biofilm only were observed on the samples covered with the studied coatings after 1 year immersion in Black sea, Yacht harbor, Nessebur.

Several parameters: total chlorophyll, carotenoids content, and biofilm dry mass were used to evaluate the formed multi-species biofilm and the results are presented in Fig. 4. The chlorophyll and carotenoids contents that are characteristics for the main algae component of the studied biofilms were very low as it is depicted in Fig. 4(a) and (b), irrespectively. For comparison, the chlorophyll content in the lives of the most plants is of about 40–70 $\mu\text{g}/\text{cm}^2$ [34] whereas that in the biofilms formed on any sample covered with the studied coatings is below 1.42 $\mu\text{g}/\text{cm}^2$. The effect of the studied NIS, NAO or combinations of NIS and NAO onto the total chlorophyll (Fig. 5a) and carotenoids content (Fig. 5b) was similar: individually, NIS and NAO significantly reduced the total chlorophyll and carotenoids content but the reduction was most pronounced at the samples with coatings containing both, NIS and NAO (Fig. 5a, b, columns 6 and 7): down to 7–8 and 10 folds, irrespectively. The dry mass of the biofilms on the coatings containing NIS, NAO or both NIS and NAO was also sharply decreased as compared to that of the biofilm on the control coating without additives, this effect most prominent at the coatings containing both, NIS and NAO, down to about 4 folds (Fig. 4c, columns 6 and 7).

It is interesting to be noted that the coatings containing NIS or NIS and NAO were very easily and totally cleaned by running water or gentle wrapping. This result as well as the data in Fig. 4 demonstrate the high potential of the relevantly selected NIS and non-toxic NAO to reduce multi-species biofilm formation individually and better in combinations: NIS to reduce the initial microbial attachment and together with NAO to inhibit microbial EPS crosslinking.

On the contrary to the relative good mono-culture bacterial adhesion, a scarce multi-species biofilm only was observed on the studied coatings after one year immersion in seawater. This could be due to different mechanisms of mono- and multi-species biofilm

formation; in the second case most probably due to a concurrent microbial attachment as a result of a concurrent microbial EPS adsorption, a phenomenon similar to the concurrent blood plasma proteins adsorption, well-known as Vroman effect.

3.6. X-Ray photoelectron spectroscopy

Having for the observed alterations of the elastic modulus, E_t of the cross-linked compositions as well as the chemistry of the evaluated additives and the possible chemical interactions with other composite components, XPS spectra were taken out of a cross-linked control coating (without additives) and those containing NIS or NAO. Detailed Si2p and O1s peaks of the coatings: control without additives (curve 1), or containing 1 wt. % NIS (curve 2) or 2 wt.% NAO (curve 3) are presented in Fig. 5. It is evident that the Si 2p peak, centered at 102.7 eV (Si–O bonds in siloxanes) [35,36] of the control coating (Fig. 5a, curve 1) is slightly shifted to the lower binding energy (at 102.4 eV) for that containing NAO (Fig. 5a, curve 3) indicating the presence of other type Si–O bonds; the O1s peak at 532.7 eV (oxygen bonded to Si) [34,35] of the control coating (Fig. 5b, curve 1) is also slightly shifted to the lower binding energy (at 532.3 eV) for the coating containing NAO (Fig. 5b, curve 3) suggesting the presence of other type oxygen to silica bonds. No shifting was observed of the C 1s peak, centered at 285 eV ($-\text{CH}_3/-\text{CH}_2-$ bonded to Si) [31,36]. The slight shifting of both, the Si 2p and O 1s peaks to the lower binding energies when NAO presents in the coating composition, indicates the presence of other Si–O bonds, formed with participation of NAO. Most probably the interaction is between its unstable OH–groups bonded to aromatic ring and $-\text{OCH}_2\text{CH}_3$ –groups of the used silane cross-linker. In this way one point attached to the vulcanization network, the NAO makes it less dense. This explains the slightly decreased elastic modulus, E_t of the vulcanized coating compositions containing NAO (Table 1). No shifts were observed for Si 2p, O 1s and C 1s of the coating containing NIS as compared to those of the control coating (without additives). That was demonstration for miss of chemical interactions of NIS.

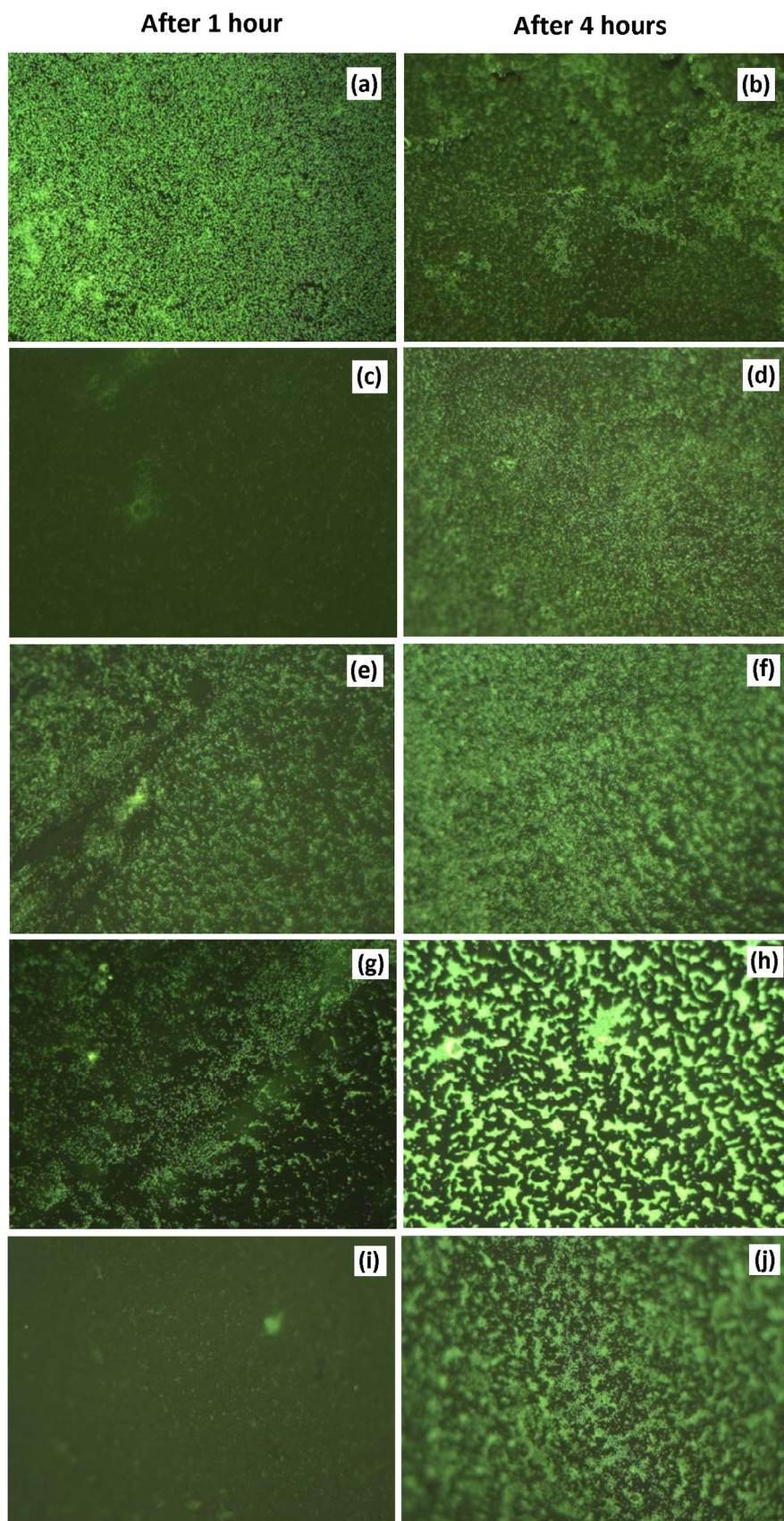


Fig. 3. Bacterial adhesion (magnification 200 $\times$) after 1 h (left column of pictures: a, c, e, g, and i) and 4 h (right column of pictures: b, d, f, h, and j) exposure in artificial sea water (ASW) on bare glass (a, b); glass covered by control coating without additives (c, d); or containing 0.5 wt.% NIS (e, f); 1.5 wt.% NAO (g, h); or 0.5 wt.% NIS and 1.5 wt.% NAO (i, j).

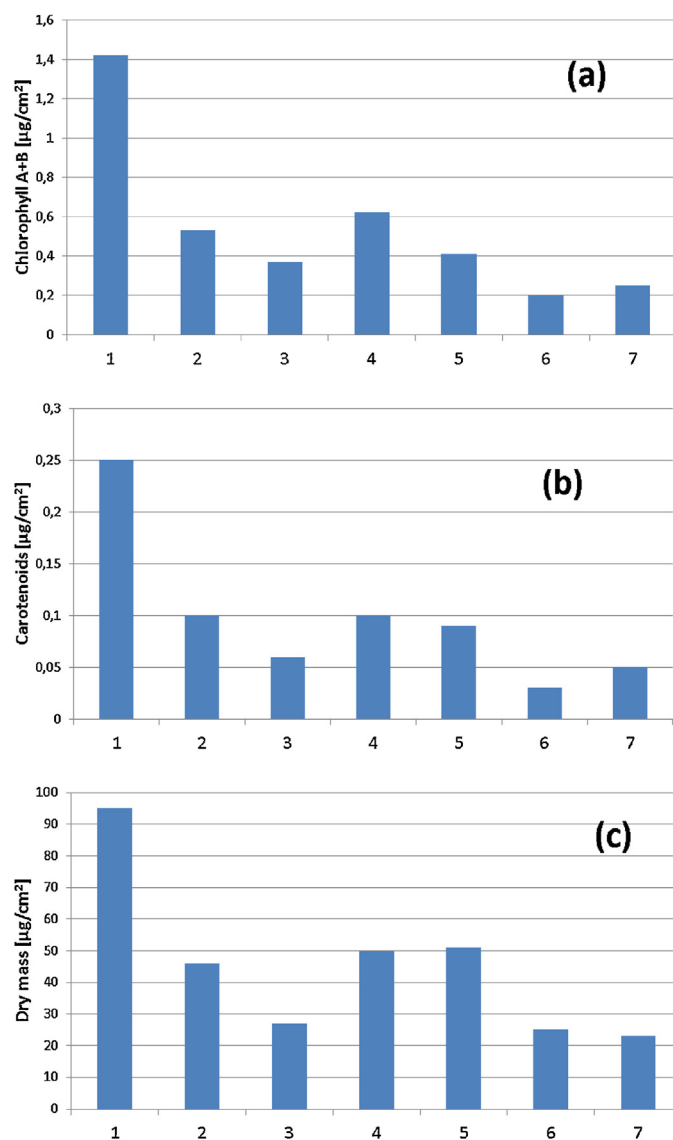


Fig. 4. Total chlorophyll (a), carotenoids content (b) and dry mass (c) of biofilm formed on coating without additives (column 1), or containing: 0.5 wt.% NIS (column 2); 1.0 wt.% NIS (column 3); 0.5 wt.% NAO (column 4); 2.0 wt.% NAO (column 5); 0.5 wt.% NIS and 1.5 wt.% NAO (column 6); 0.5 wt.% NIS and 1.5 wt.% NAO (column 7).

4. Discussion

All experimental data presented so far demonstrated the superior efficiency of relevantly selected NIS, NAO and their combinations to reduce mussel adhesion (down to approximately zero), algae growth (down to approximately zero), and marine biofouling (macrofouling down to zero; multi-species biofilm down to 10 fold) of biocides-free, low surface energy, siloxane antifouling coatings. Their improved anti-biofouling properties could be ascribed to a “down up” formation of a thin, plane, low surface energy, “bioactive” layer, as proved by AFM, contact angle measurement, Depth Sensing Indentation (DSI), bioadhesion and marine biofouling testing. The “bioactivity” was shown to result from biofilm dispersal activation and oxidative cross-linking of microbial and macrofoulers inhibition by NIS and NAO.

The proved by XPS analyses chemical interaction between NAO and the silane crosslinker was shown to result in a decreased elastic modulus, Et contributing also to the improved antifouling properties of the studied coatings. The monoculture bacterial adhesion was less influenced by NIS and NAO whereas the multi-species marine biofilm formation was strongly reduced (down to 10 fold)

and the developed scare biofilm was totally and easily removed by running water or gentle wrapping. This difference can be explained by the versatile nature of the microbial EPS, their concurrent adsorption and respectively, concurrent microbial attachment, in the second case.

5. Conclusions

Relevantly selected non-ionic surfactant and non-toxic antioxidant exhibited strong antibiofouling activity in biocides-free low adhesive polysiloxane antifouling composition coatings, the effect being most prominent as expected at those containing both non-ionic surfactant and non-toxic antioxidant.

The coatings demonstrated superior antifouling properties in laboratory and marine conditions: marine macro-fouling, mussel adhesion and algae growth all reduced to zero; multi-species biofilm reduced sharply, 6–10 fold; as well as total remove of the formed scare biofilm by gentle cleaning out with running water or wiping.

“Down up” formed thin, plane “bioactive” surface layer was proved and characterized (by Depth Sensing Indentation, AFM,

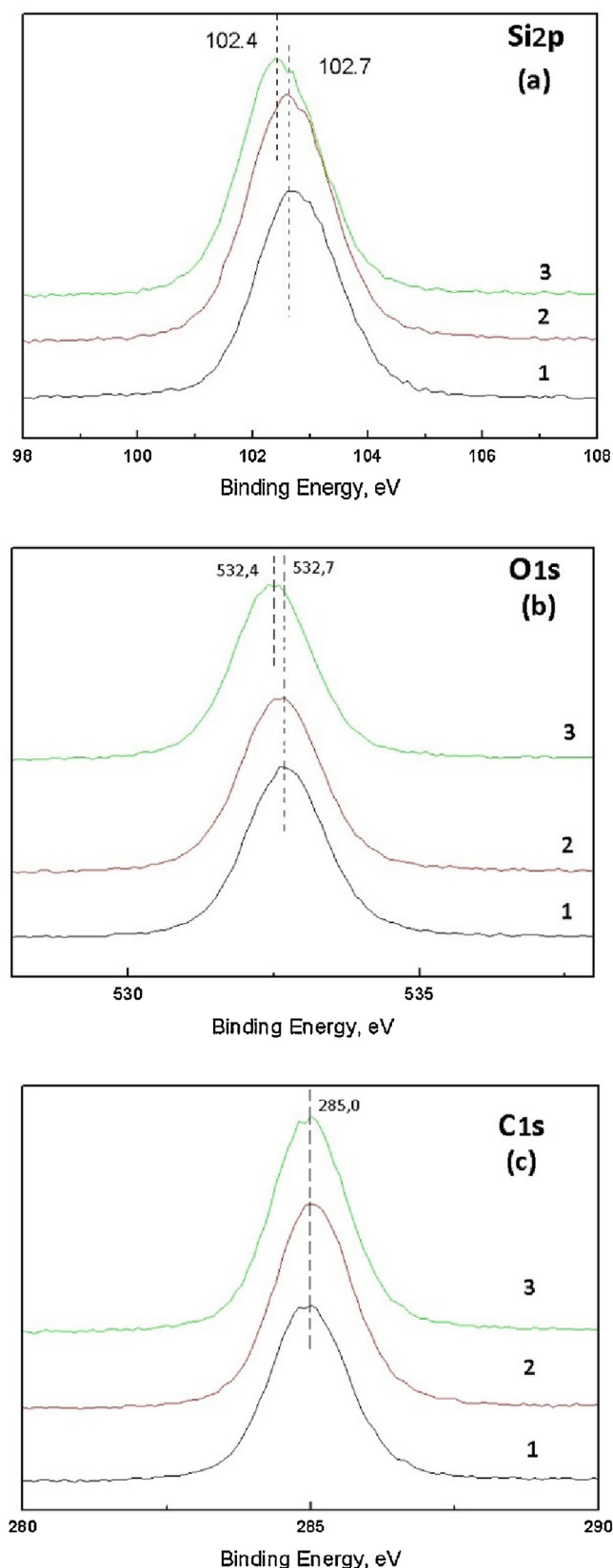


Fig. 5. Si 2p (a), O 1s (b) and C 1s (c) XPS peaks of cross-linked coatings: control, without additives (curve 1) and containing 1.0 wt.% NIS (curve 2) or 2.0 wt.% NAO (curve 3).

contact angle measurement and XPS) in which is supposed the non-ionic surfactant to impede the initial microbial adhesion, together with the non-toxic antioxidant to inhibits the oxidative crosslinking of EPS and later to assists the biofilm dispersal.

Contrary to the good monoculture bacterial adhesion, scare multi-species marine biofilm was formed on the studied coatings, this fact explained by a different impact of the used surfactant and antioxidant on the adsorption and crosslinking of secreted by one marinobacter and a variety of microbe EPS for those conventional is a concurrent adsorption.

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